# Team Details

**Team Name:** BYTE BRIGADE

| SR. NO | ROLE | NAME | ACADEMIC YEAR |
|---|---|---|---|
| 1 | **Team Leader** | *Prakhar Bajaj* | *2nd Year* |
| 2 | Member 1 | *Raghav Soni* | *2nd Year* |
| 3 | Member 2 | *Parth Kumar* | *2nd Year* |
| 4 | Member 3 | *Mayank Lodhi* | *2nd Year* |

***Problem Statement PS01 (KLA) - AI-Based Restoration of Degraded Images for Semiconductor Inspection***


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# Problem Statement Addressed

## PS01 - KLA: AI-Based Restoration of Degraded Images for Semiconductor Inspection

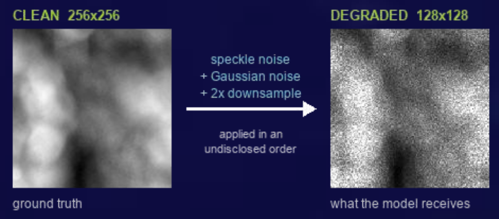
**DESCRIPTION / DETAILS**

*Wafer inspection images arrive noisy and low-resolution: fast, low-dose scanning trades image quality for throughput.*

*Three degradations are present - speckle noise (scales with brightness), additive Gaussian sensor noise, and 2x downsampling - applied in an undisclosed order.*

*Real defects hide inside this noise. A missed defect can cost an entire wafer; a false alarm wastes engineer time.*

*Task: learn a mapping from a 128x128 degraded image to its clean 256x256 original, scored on PSNR, SSIM, LPIPS and end-to-end speed.*



# Idea Description

One small network restores a noisy 128x128 image to a clean 256x256 one - denoising and 2x super-resolution in a single pass, trained only on the data provided.

**KEY CONCEPT & APPROACH**

*One small network denoises and 2x super-resolves in a single pass.*

*Supervised on the 3,200 provided pairs - no external data, no pretrained restoration weights.*

*Predicts only a correction on a bicubic upscale, which strongly constrains hallucination.*

**SOLUTION OVERVIEW**

*NAFNet-SR: 0.53 M parameters, 2.3 MB checkpoint - runs on a laptop CPU.*

*Trained on the judged metrics: Charbonnier + SSIM + LPIPS loss.*

*28.00 dB PSNR / 0.759 SSIM / 0.175 LPIPS - beats bicubic on all three.*

*End-to-end 25 ms per image on a T4 GPU (40 images/second).*

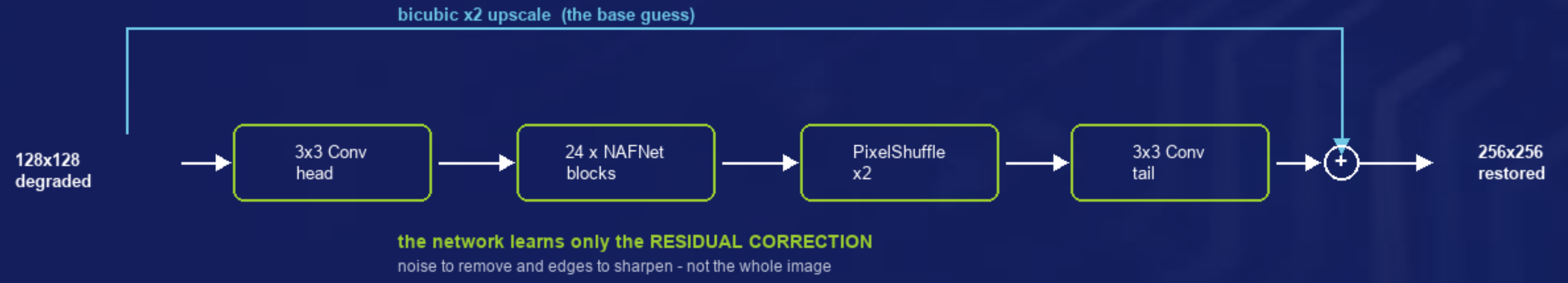
# Proposed Solution

NAFNet-SR: 24 simplified restoration blocks and a learned 2x upsampler, added to a bicubic upscale so the network predicts only a correction.

SOLUTION DETAILS

*WHY NAFNet - SOTA quality, no transformer, no activation function.   LOSS - Charbonnier + 0.15 SSIM + 0.05 LPIPS (no GAN loss).*

*TRAINING - 3,000/100/100 seeded split, 128 px crops, batch 16, 100 epochs, ~2 h on a free Colab T4.   INFERENCE - directories in/out, batched, tiled.*



# Innovation and Uniqueness

We train directly on the judged metrics, and constrain the model to correct rather than generate - decisive when invented structure could be mistaken for a defect.

**KEY INNOVATION**

*We train directly on the scoreboard: SSIM and LPIPS terms sit inside the loss, so the model optimises two of the three judged metrics rather than hoping they follow from pixel accuracy.*

*Hallucination is strongly constrained: the global bicubic skip means the network predicts a correction rather than generating an image freely.*

*Float-faithful pipeline: degraded inputs legitimately exceed [0,1] and we never clip them; clipping is applied only to outputs, where ground truth guarantees the range.*

**COMPETITIVE ADVANTAGE**

*SMALLER: 0.53 M parameters (2.3 MB) versus 10-26 M for Restormer or SwinIR, so throughput - an explicit scoring axis - is a strength.*

*FASTER: 25 ms per image end-to-end on a T4; all 400 test images in ~10 s.*

*TRUSTWORTHY: beats the bicubic baseline on 100 of 100 held-out images; failure cases are published and analysed rather than hidden.*

*REPRODUCIBLE: seeded splits, one-command training and inference, weights and runtime report committed to the repository.*

# Impact and Benefits

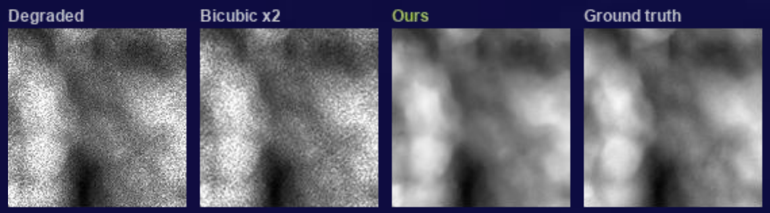
Better on all three judged image-quality metrics, at 25 ms per image - recovering detail that noise would otherwise hide, with no new hardware.

### Primary Impact

Detail hidden inside sensor noise becomes visible again - fewer missed defects and fewer false alarms.
Enables faster or lower-dose scanning: recover quality in software, so existing microscopes inspect more wafers.

### Quantifiable Outcomes

PSNR  22.96 -> 28.00 dB   (+5.04 dB)
SSIM  0.537 -> 0.759   |   LPIPS  0.433 -> 0.175
25 ms/image - all 400 test images in ~10 s
0.53 M parameters, 2.3 MB checkpoint - runs on CPU




# Technology & Feasibility/Methodology Used

PyTorch and NAFNet-SR, trained in about two hours on a free Colab T4, and deployable on either a GPU or a plain laptop CPU.

IMPLEMENTATION STRATEGY

*Trained end to end on a free Google Colab T4 in about two hours; no paid infrastructure was used at any stage.*
*Inference runs on GPU or CPU. Batched on CUDA, tiled for large images, and verified on 128, 256 and non-square inputs.*
*Deployment is a drop-in folder-to-folder script plus a Streamlit UI for operators; evaluators need no source-code edits.*
*Measured, not estimated: benchmark.py reports end-to-end runtime with hardware and software versions stated.*

PyTorch 2.11 - NAFNet-SR

NVIDIA T4 / H100 - CPU capable

Colab - Git - Streamlit

# GitHub & Video Link

### GitHub Repository

github.com/prakharbajaj1551/semicon-image-restoration

### Prototype / Simulation Video

https://www.youtube.com/watch?v=RMSDaviTOIw

# Research and References

## Research Background & Methodology

Every component is drawn from peer-reviewed restoration research, not invented for this hackathon.

*Built on published, peer-reviewed research rather than an unproven design: NAFNet (ECCV 2022) matches transformer restorers at a fraction of the cost.*
*Loss follows restoration literature - Charbonnier for pixel fidelity, SSIM for structure, LPIPS for perceptual quality.*
*Validated on a held-out split with no selection leakage, against a bicubic baseline, with failure cases analysed.*

## References & Citations

The three papers behind our architecture, our upsampler and our perceptual metric.


Ref 1: Chen et al., Simple Baselines for Image Restoration (NAFNet), ECCV 2022 - arxiv.org/abs/2204.04676

Ref 2: Zhang et al., The Unreasonable Effectiveness of Deep Features as a Perceptual Metric (LPIPS), CVPR 2018

Ref 3: Shi et al., Real-Time Single Image Super-Resolution Using an Efficient Sub-Pixel CNN, CVPR 2016